03 · Evangelical density and recorded domestic violence in Brazilian municipalities

Final report generated by the pipeline (python run_all.py). Tables and figures are the pipeline's own outputs; the text comes from the paper's README.

Question

Do municipalities with a larger share of evangelicals record more (or less) domestic violence against women?

Result

• With state fixed effects and HC1 errors, the evangelical-density coefficient is 0.0009 (SE 0.0009, p = 0.325), on 5046 municipalities.

• Adding the non-domestic notification proxy raises R^2 from 0.171 to 0.439; the proxy coefficient is 0.4237, significant at 1%, and evangelical density stays at 0.0009.

• The Poisson model with a population offset gives 0.0080, an incidence-rate ratio of 1.0081, not significant.

• In WLS the coefficient is 0.0019, not significant.

• Across the six robustness columns, the coefficient reaches significance only once: at 10%, with errors clustered by state (0.0009*).

• With the 3-year denominator the coefficient is 0.0007 (p = 0.414).

Reading: no robust association. What explains recorded rates is how much a municipality notifies, not its religious composition. Associational, not causal; the denominator is the whole population, not women at risk.

Data checks

14 pass, 1 warnings, 0 fail (details in output/data_checks.json).

Main tables

Main models: evangelical density and recorded domestic violence

	(1)	(2)	(3)	(4)	(5)
Dependente	VD	VD	log(1+VD)	VD	VD
Estimador	MQO	MQO	MQO	MQO	WLS
Densidade evangélica	0.0009	0.0013	0.0007	0.0009	0.0019
	(0.0009)	(0.0023)	(0.0005)	(0.0008)	(0.0022)
Densidade evangélica^2/100		-0.0009			
		(0.0047)			
log(Renda pc)	0.0886***	0.0883***	0.0515***	0.0338***	0.0859***
	(0.0142)	(0.0143)	(0.0078)	(0.0112)	(0.0263)
log(População)	-0.0430***	-0.0432***	-0.0258***	-0.0248***	-0.0381**
	(0.0080)	(0.0081)	(0.0040)	(0.0070)	(0.0166)
Notif. não-domésticas				0.4237***	
				(0.0279)	
Efeitos fixos de UF	Sim	Sim	Sim	Sim	Sim
Observações	5046	5046	5046	5046	5046
R^2	0.171	0.171	0.229	0.439	0.122

Source: output/tab_principal.tex

Robustness: evangelical density coefficient across samples and errors

	Base 2011–13	2011–13 (comum)	2013–15 (comum)	Sem capitais	Cluster UF	Pop. >= 5k
Densidade evangélica	0.0009	0.0008	0.0016	0.0009	0.0009*	-0.0010
	(0.0009)	(0.0009)	(0.0012)	(0.0009)	(0.0005)	(0.0009)
Controles+EF UF	Sim	Sim	Sim	Sim	Sim	Sim
Observações	5046	4972	4972	5019	5046	3973
R^2	0.171	0.170	0.278	0.171	0.171	0.163

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Poisson with population offset

	Coef.	IRR
Densidade evangélica	0.0080	1.0081
	(0.0077)	
log(Renda pc)	0.2924***	1.3396
	(0.0846)	
Offset log(pop.)	Sim	
EF de UF	Sim	
Observações	5046	

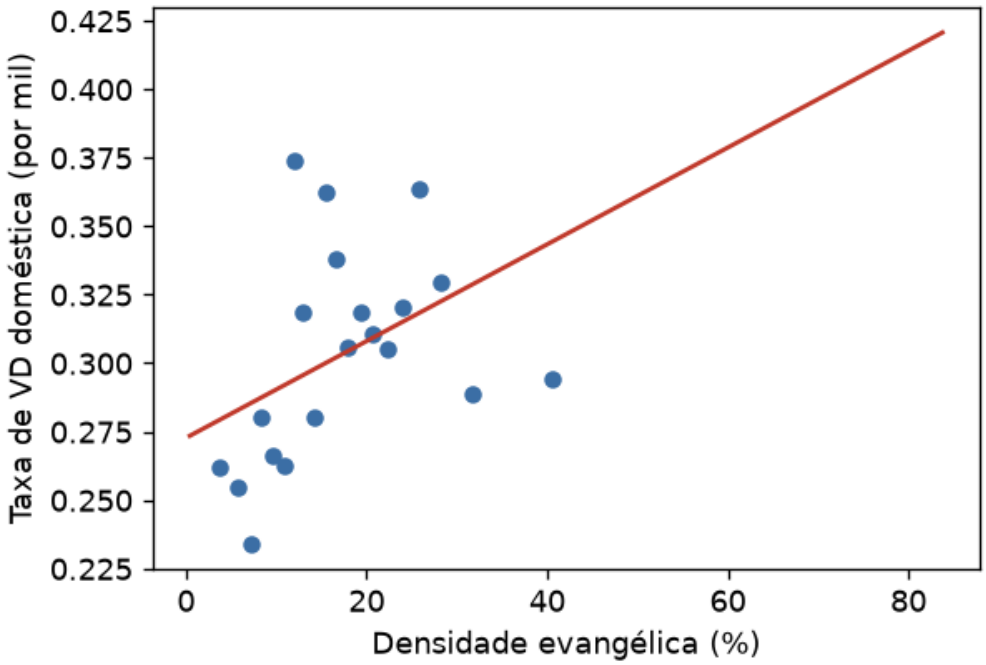
Source: output/tab_poisson.tex

Refinement: rate over the full 3-year window

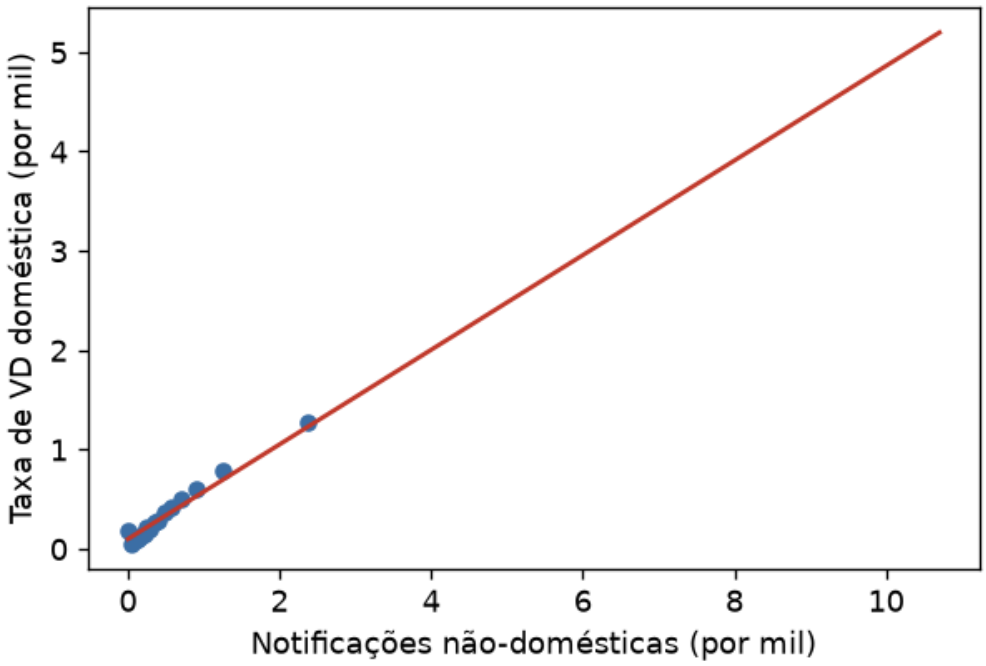
	(1) as submitted	(1) 3-year denominator	(4) as submitted	(4) 3-year denominator
Densidade evangélica	0.0009	0.0007	0.0009	0.0006
	(0.0009)	(0.0009)	(0.0008)	(0.0007)
Notif. não-domésticas			0.4237***	0.4528***
Observações	5046	5046	5046	5046
R^2	0.171	0.153	0.439	0.481

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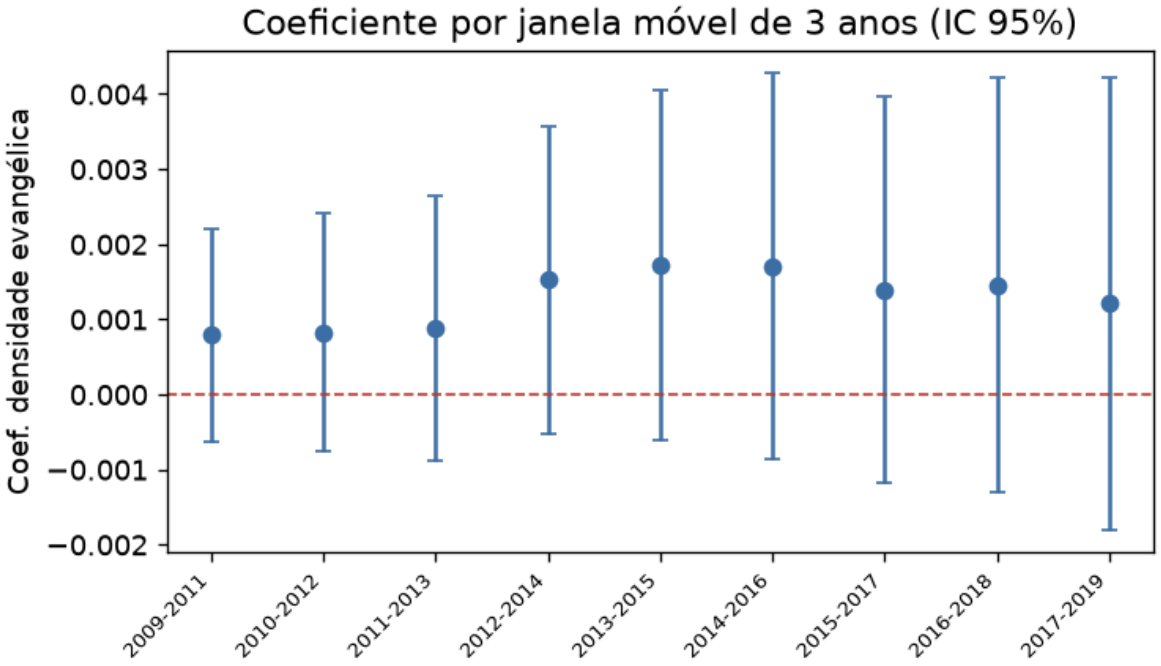
Main figures



Source: output/figures/fig_evang_vd.png



Source: output/figures/fig_proxy_vd.png



Source: output/figures/fig_coef_janelas.png